DSF521 Assignment4

Due on May 3, 2025

1 Exercise 5.1

Show that the scalar SDE

$$\begin{aligned} dX_t &= \alpha X_t dt + \sigma dW_t, \\ X_0 &= x_0, \end{aligned}$$

has the solution

$$X_t = e^{\alpha t} \cdot X_0 + \sigma \int_0^t e^{\alpha(t-s)} dW_s \tag{1}$$

by differentiating X as defined by eqn (1) and showing that X so defined actually satisfies the SDE. Hint: Write eqn (1) as

$$X_t = Y_t + Z_t \cdot R_t,$$

where

$$\begin{aligned} Y_t &= e^{\alpha t} \cdot x_0, \\ Z_t &= e^{\alpha t} \cdot \sigma, \\ R_t &= \int_0^t e^{-\alpha s} dW_s, \end{aligned}$$

and first compute the differentials dZ, dY and dR. Then use the multidimensional Itô formula on the function $f(y, z, r) = y + z \cdot r$.

2 Exercise 5.5

Suppose that X satisfies the SDE

$$dX_t = \alpha X_t dt + \sigma X_t dW_t.$$

Now define Y by $Y_t = X_t^\beta$, where β is a real number. Then Y is also a GBM process. Compute dY and find out which SDE Y satisfies.

3 Exercise 5.9

Use a stochastic representation result in order to solve the following boundary value problem in the domain $[0,T] \times R$.

$$\frac{\partial F}{\partial t} + \mu(t,x)\frac{\partial F}{\partial x} + \frac{1}{2}\sigma^2(t,x)\frac{\partial^2 F}{\partial x^2} + k(t,x) = 0,$$
$$F(T,x) = \Phi(x).$$

Here μ, σ, k and Φ are assumed to be known functions. Prove that this problem has the stochastic representation formula

$$F(t,x) = E_{t,x}[\Phi(X_T)] + \int_t^T E_{t,x}[k(s,X_s)]ds,$$

where as usual X has the dynamics

$$dX_s = \mu(s,X_s)ds + \sigma(s,X_s)dW_s,$$
$$X_t = x.$$

Hint: Define X as above, assume that F actually solves the PDE and consider the process $Z_s = F(s,X_s)$.

4 Exercise 7.3 (Modified)

Consider the standard Black–Scholes model. Derive the Black–Scholes formula for the European **put** option.

5 Exercise 7.4

Consider the standard Black–Scholes model. Derive the arbitrage free price process for the T-claim $\mathcal{X}$ where $\mathcal{X}$ is given by $\mathcal{X} = S(T)^{\beta}$. Here β is a known constant.
Hint: For this problem you may find Exercises 5.5 and 4.4 useful.